Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Geological Society of America Bulletin

Magnetic stratigraphy applied to synsedimentary slumps, turbidites, and basin analysis: The Scaglia Limestone at Furlo (Italy)

WALTER ALVAREZ and WILLIAM LOWRIE

Geological Society of America Bulletin 1984;95;324-336
doi: 10.1130/0016-7606(1984)95<324:MSATSS>2.0.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Bulletin

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Magnetic stratigraphy applied to synsedimentary slumps, turbidites, and basin analysis: The Scaglia Limestone at Furlo (Italy)

WALTER ALVAREZ *Department of Geology and Geophysics, University of California, Berkeley, California 94720*
WILLIAM LOWRIE *Institut für Geophysik, ETH-Hönggerberg, CH-8093 Zürich, Switzerland*

ABSTRACT

The previously studied Italian Scaglia sections at Gubbio and Moria consist entirely of pelagic limestone and marl. By contrast, the Furlo section is characterized by synsedimentary slump folds and slide masses in the lower portion (Cenomanian-Coniacian) and by intercalation of white calcarenitic turbidites in the upper part (Campanian-Paleocene). Demagnetization to 20 mT and 350 °C or higher reveals a single, stable direction after removal of an initial low-coercivity component. Despite the stratigraphic complications, the pattern of polarity zones clearly matches Gubbio and the sea-floor anomalies. The abrupt faunal break at the end of the Cretaceous was located, and foram zonation shows that the very base of the Paleocene is missing. Sedimentation-rate determinations based on the magnetic polarity zonation have clarified the origin of stratigraphic complications in the Furlo Paleocene. Most samples were taken from pelagic beds, but ten samples from calcarenitic turbidites yielded directions indistinguishable from those of adjacent pelagic beds. Detailed sampling of a turbidite 33 cm thick confirmed that this lithology is an accurate recorder of the geomagnetic field. Synsedimentary slumps range from coherent slide blocks with undeformed bedding to smoothly curving folds indicating flow of very soft sediment. Coherent slide blocks show declinations rotated by tens of degrees away from the typical regional direction, but their inclinations are unaffected. Soft-sediment slump folds yield typical regional directions, indicating resetting of the remanent magnetization after deformation. Using stratigraphic, structural, and paleomagnetic data, it is possible to identify, in the lower part of the section, a basal nonslumped unit overlain by seven slump units with various characteristics, interlayered with four units of pelagic sediments which were deposited on older slump units but which themselves were not affected by synsedimentary slumping. This paper emphasizes the value of paleomagnetic stratigraphy as a tool in detailed studies of sedimentology, stratigraphy, and basin analysis.

INTRODUCTION

The Cretaceous and lower Tertiary pelagic limestones and marls of the Umbrian and Marches Apennines have been an extremely fruitful subject for paleomagnetic study (Lowrie, Channell, and Alvarez, 1980a). This work has focused on three main topics: (1) the history of rotations and latitude changes of the underlying Italian continental crust, especially its role as part of the "Adriatic Promontory" of Africa (Lowrie and Alvarez, 1974, 1975; Channell and Tarling, 1975; Channell and Horvath, 1976; Channell, 1976; VandenBerg and others, 1978; Channell and others, 1979); (2) bending of the Umbria-Marches Apennines during formation of folds and thrusts (Lowrie and Alvarez, 1975; Channell and others, 1978); and (3) stratigraphic studies that have tied the magnetic reversal time scale to the micropaleontological zonation over almost 100 m.y. from the Early Cretaceous to the base of the Miocene (Premoli Silva and others, 1974; Lowrie and Alvarez, 1977a, 1977b, 1981; Roggenthen and Napoleone, 1977; Alvarez and others, 1977; Alvarez and Lowrie, 1978; Lowrie and others, 1980a, 1980b, 1982; Napoleone, 1980; Premoli Silva and others, 1980; Napoleone and others, 1983).

The present paper carries the magnetostratigraphic work in a new direction. Previous studies have concentrated on long, continuously pelagic sections which were expected to have preserved the most complete stratigraphic record, essentially free of depositional hiatuses. We now report on three sections from a nearby area where much of the Upper Cretaceous is represented by syndepositional slumps and slide blocks, with redeposited limestone turbidites common in the uppermost Cretaceous and Paleocene. (In this paper, the term "slump" will be used both for bodies deformed by folding while the sediment was soft and for coherent, syndepositional slide masses.)

The area studied is located on the southwest flank of the Furlo anticline, where the Via Flaminia traverses the gorge cut by the Candigliano River (Fig. 1). Three sections (Furlo Flaminia, Furlo Upper Road, and Pietralata) were studied in order to piece together a long continuous section.

The Furlo sections were selected for study because they show notable differences from the Bottaccione section at Gubbio, which has been the subject of detailed lithostratigraphic, sedimentological, paleontologic, paleomagnetic, and geochemical studies. Whereas the Bottaccione section records essentially undisturbed deposition of pelagic limestones and marls through the entire time interval represented by the middle Cretaceous to middle Eocene Scaglia Rossa, the Furlo Scaglia Rossa is marked by extensive synsedimentary slumping and by the presence of numerous white beds of unmistakable turbidite origin.

In this study, we have obtained positive answers to the following questions: Do fine-grained limestone turbidites record the geomagnetic field as faithfully as do fine-grained pelagic limestones? Is the distinctive Late Cretaceous–Paleocene magnetic polarity sequence recognizable in a section containing turbidites, without serious distortion by turbidity-current erosion and episodic depositional events? Is it possible to locate foraminiferal-zone boundary marker events, in particular, the Cretaceous-Tertiary boundary, in nonturbidite beds within a turbidite-bearing sequence? Can paleomagnetism supply information useful for understanding synsedimentary slumping?

LITHOSTRATIGRAPHY AND BIOSTRATIGRAPHY

Here we give only a brief description of the principal lithostratigraphic differences between the Furlo and Bottaccione sections. The Furlo anticline is abnormal in having very thin Majolica and Fucoid Marls (Lower Cretaceous). The Scaglia Bianca (Cenomanian) is of normal thickness and lithology (Montanari, 1979, Fig. 40). The Bonarelli Level, a 1-m interval of dark, siliceous, carbonaceous shale, which we take as the boundary between the Scaglia Bianca and the Scaglia Rossa, occurs as three horizons, rather than the normal single horizon (Fig. 2).

Geological Society of America Bulletin, v. 95, p. 324–336, 11 figs., 2 tables, March 1984.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

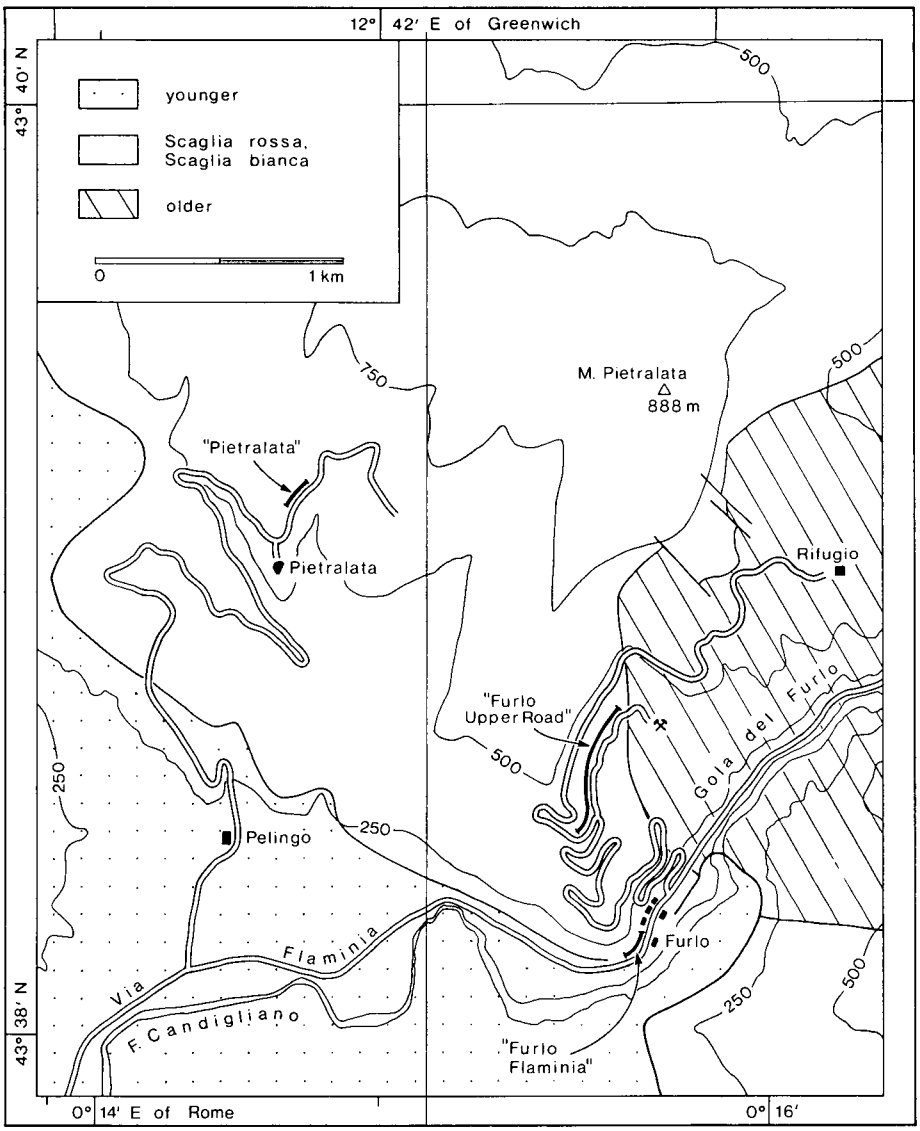


Figure 1. Location of the three sections studied, on the southwest flank of the Furlo anticline, about 200 km north of Rome and 120 km east of Florence. The area shown is located 35 km north-northeast of Gubbio.

This repetition is due to synsedimentary slumping; minor stratigraphic details within and adjacent to Bonarelli I are repeated identically in Bonarelli II, whereas Bonarelli III is a thin smear of deformed material. Beginning between Bonarelli Levels I and II, and continuing upward for 185 m, various intervals in the section are marked by disturbance of the bedding due to slumping, in some cases probably augmented by later tectonic deformation. The presence of slump structures ranging from smoothly curving folds to blocky sheared bedding shows that the early deformation affected sediments in various states of induration and stiffness. The lower cherty member of the Scaglia Rossa is abnormally thick in the Furlo section (150 m, instead of the usual 100 m). Whereas the foram content of the Scaglia Rossa limestones in the Bottaccione section is nearly constant at about 10%, the nonturbidite beds of the Furlo Scaglia Rossa range from nearly pure micrite to >70% forams (A. Montanari, 1980, personal commun.).

The middle, chert-free member of the Scaglia Rossa at Furlo includes about 75 white limestone beds, 10 to 100 cm thick, that show unmistakable signs of origin as turbidites, notably Bouma sequences and sole markings (Montanari, 1979; Wezel, 1979). Montanari (1979) noted the rhythmic occurrence of the turbidite beds at Furlo, with two maxima of thickness and abundance. The turbidite beds are distinct from the intervening pink beds, which lack turbidite features and which in this part of the section have high foram contents. The turbidite sequences are almost identical between the Flaminia and Upper Road sequences, which are 600 m apart; between the Upper Road and Pietralata sections, 1,800 m apart, there are substantial changes in the pattern of turbidites, and it is not possible to make unambiguous correlations.

A. Micarelli (1975, personal commun.) and F. Medizza (1977, personal commun.) made foraminiferal determinations on some of our early samples and found that the pink, non-turbidite beds do preserve a reliable sequence of foraminiferal assemblages. Subsequently we located the Cretaceous-Tertiary boundary and found that the characteristically porcellaneous lithology of the basal Paleocene *Globigerina eugubina* zone is missing, although microfossils of this thin zone may be present. More recently, A. Montanari (1980, personal commun.), made an approximate foraminiferal zonation on thin sections from the Pietralata Paleocene, and I. Premoli Silva (1981, personal commun.) identified isolated specimens from three critical levels at Pietralata.

MAGNETIC PROPERTIES

The magnetic properties of the pink Scaglia Rossa and the white Scaglia Bianca limestones have been described elsewhere in detail (Channell and others, 1982; Lowrie and Heller, 1982) and will be only briefly summarized here. Except in the red Paleocene Scaglia Rossa, the remanences are comparatively uncomplicated.

In the white Scaglia Bianca, the natural remanent magnetization (NRM) is carried by magnetite; a minor high-coercivity component (possibly hematite or goethite) is often present. The remanent magnetization is thought to have been acquired soon after deposition by statistical alignment of the fine-grained magnetite in interstitial voids in the water-logged sediment, akin to the postdepositional remanence found in many other sediments (Irving, 1957; Irving and Major, 1964; Kent, 1973; Barton and others, 1980; Hamano, 1980; Tucker, 1980).

The hematite that gives pink and red Scaglia Rossa samples their color forms diagenetically, possibly from a goethite precursor (Channell and others, 1982). The proportion of the NRM carried by hematite and the timing of its growth are important factors in determining the Scaglia Rossa magnetic properties. Although the hematite component in Scaglia Rossa samples from Sicily apparently originated long after sediment deposition (Channell, 1978), the hematite component in the Cretaceous Umbrian Scaglia Rossa probably developed and acquired mag-

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

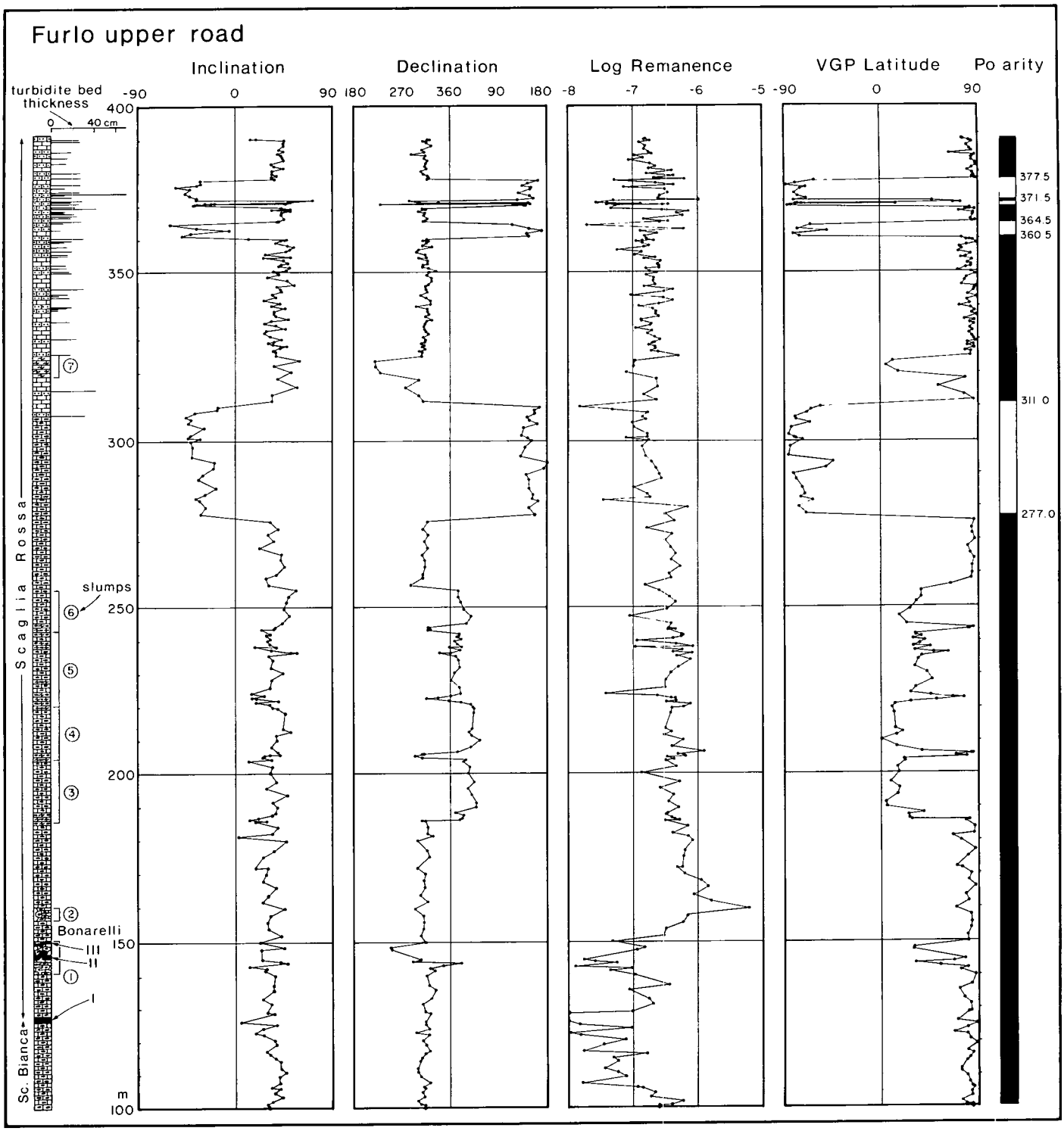


Figure 2. Magnetic results from the Furlo Upper Road section (for lithological symbols, see Fig. 5). Remanent intensity in Gauss. Slump levels (except no. 2) are reflected in the record of declinations but not of inclinations.

netization during early diagenesis (Lowrie and Alvarez, 1975; Channell and others, 1982).

The characteristic remanent magnetization (ChRM) of pink Scaglia Rossa and white Scaglia Bianca samples is carried predominantly by magnetite and could be isolated by progressive alternating-field (AF) or thermal demagnetization. After removal of an initial unstable component of low coercivity in fields of as much as 20 mT, AF demagnetization defines a vector that decays linearly to the origin of a vector diagram (Fig. 3a). Similar characteristics are revealed during thermal demagnetization, which isolates the ChRM direction in temperatures above 350 °C (Fig. 3b). The separate demagnetization techniques define the same direction in twin samples from the same core; the thermal demagnetization also reveals that only a small

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

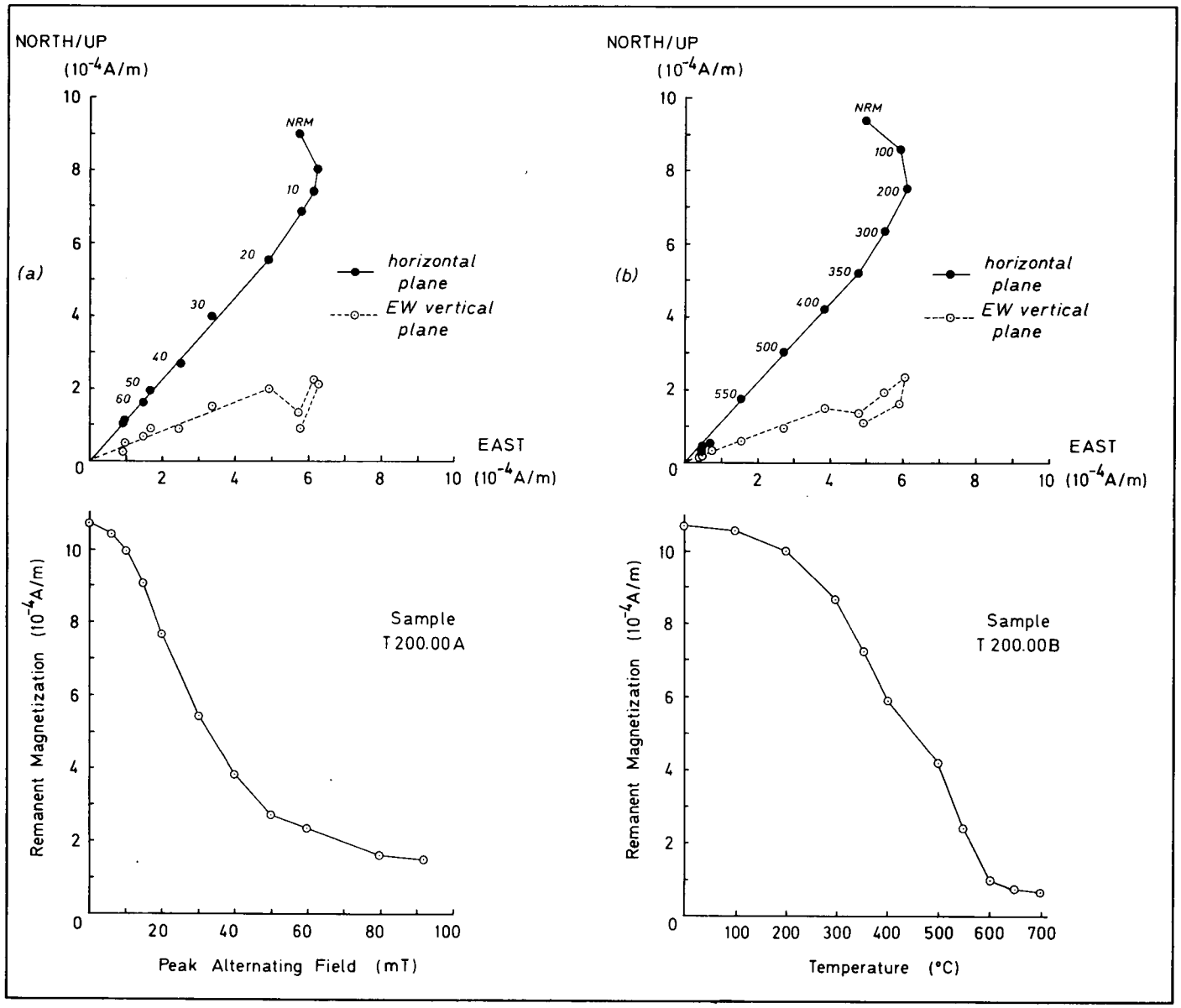


Figure 3. Results of magnetic cleaning of two samples from a single core (200-m level in the Furlo Upper Road section). a. Alternating-field demagnetization. b. Thermal demagnetization. Both techniques define the same characteristic remanent magnetization vector, and thermal demagnetization shows that magnetite is the principal carrier of remanence.

component of the NRM survives heating beyond the Curie point of magnetite (578 °C).

In Upper Cretaceous and Eocene Scaglia Rossa samples, the reddish hue is darker and the hematite is correspondingly more important. In most samples, the ChRM can be isolated by heating above 235 °C (Fig. 4; Eocene sample P73.90B and Cretaceous sample T291.80B). Many samples show partially overlapping blocking-temperature spectra, however, and the ChRM vector to the origin is defined only above 535 °C (Fig. 4; Cretaceous samples P33.68B and T279.96B). The component carried by hematite alone, at temperatures of 590 °C and above, was usually parallel to the component between the temperature at which unstable components were removed and the Curie temperature of the magnetite component (for example, sample T291.80B).

All pink Cretaceous Scaglia samples from the Pietralata and Furlo Upper Road sections were demagnetized at several stages until a stable direction was reached. AF demagnetization at 5, 10, 15, and 20 mT was followed by thermal demagnetization at 350 °C; some samples were also heated at 400 and 500 °C.

Detailed investigation of the magnetization of red Scaglia Rossa samples of Paleocene age has shown that it is multicomponent and generally consists of (1) a soft viscous component, (2) a more resistant component antiparallel to the present field's direction, (3) a composite component due to detrital magnetite and fine-grained authigenic hematite, and (4) a less fine-grained hematite component (Channell and others, 1982). The identification of the ChRM direction in red Paleocene samples from the Pietralata section was in many cases obscured by the overlap of the blocking-temperature spectra of these multiple components. The same combination of AF and thermal demagnetization stages used for pink samples, was less effective in the red samples, and further detailed demagnetizations at high temperatures were often necessary. An additional complication which

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

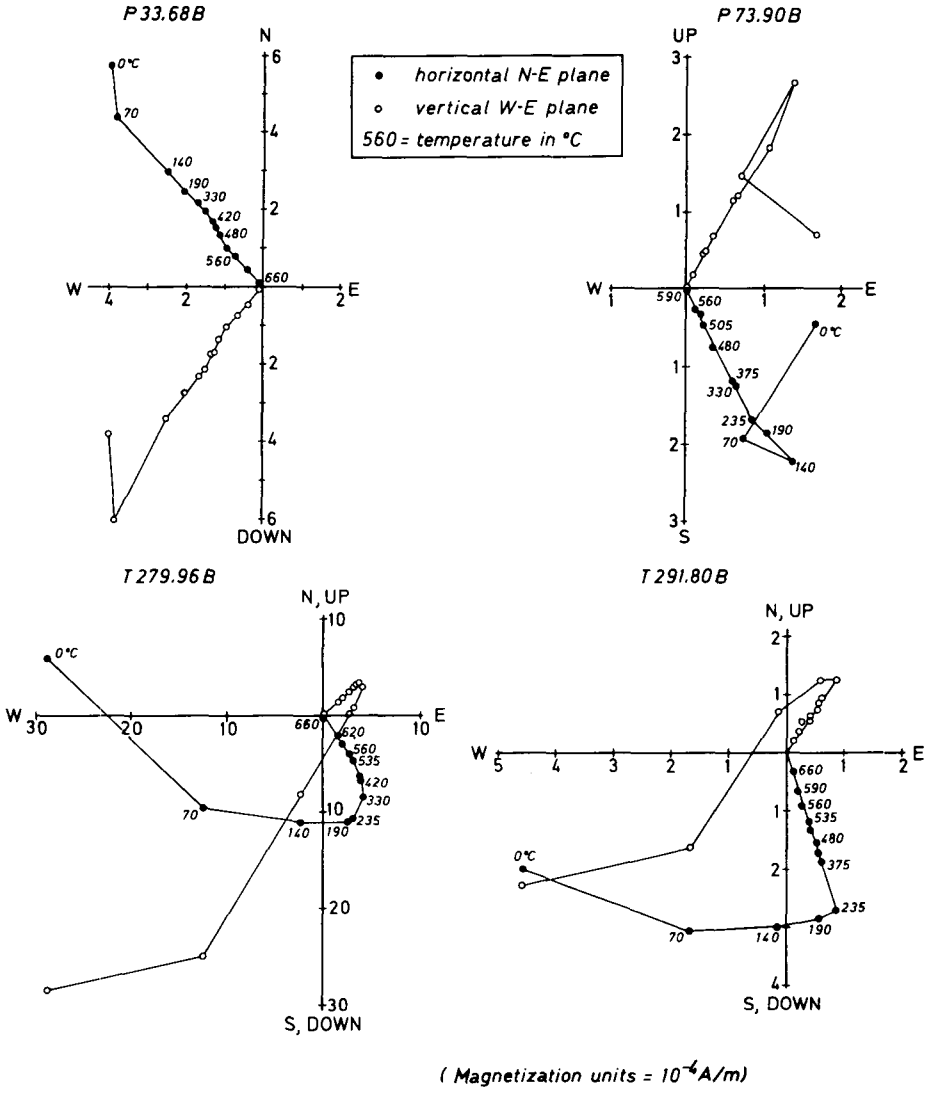


Figure 4. Thermal demagnetization of selected red Scaglia Rossa samples. The ChRM vector to the origin is defined only above 535 °C in some samples (P33.68B, T279.96B) because of overlapping blocking-temperature spectra. In other samples (for example, P73.90B), a linear vector to the origin was defined above 235 °C, and the vectors carried by the magnetite component below 560 °C and the hematite component above 590 °C were parallel (T291.80B).

then arose was the creation of new magnetite at high temperature from the red hematite pigment. This resulted in a dramatic increase in susceptibility after heating above 600 °C (Lowrie and Heller, 1982) and necessitated very good field cancellation to avoid introducing spurious thermoremanent magnetization components.

MAGNETIC POLARITY STRATIGRAPHY

The characteristic remanent magnetization direction was determined for 16 preliminary oriented block samples from the Furlo Flaminia section, for 263 drilled samples from the Furlo Upper Road section, and for 171 drilled samples from the Pietralata section. Figures 2, 5, and 6 show declinations and inclinations of ChRM directions in all acceptable drilled samples, as well as the remanent intensity at the demagnetization step selected as giving the ChRM direction.

The slumped zones are readily identifiable and are omitted from statistical analysis (Fisher, 1953) of the ChRM directions in each section (Table 1). After correcting for bedding attitude, the ChRM directions are equivalent in declination and inclination to those found elsewhere in Umbrian Scaglia Rossa sections (Lowrie and Alvarez, 1977b; VandenBerg and others, 1978; Alvarez and Lowrie, 1978). The inclinations are compatible with a latitude of deposition on the southern margin of the Tethys Ocean. The northwest-southeast declination concentrations reflect counterclockwise tectonic rotation of the Umbrian sequence, either as part of an Adriatic promontory of the African continent (Channell and Horvath, 1976; Channell and others, 1979) or as a Mediterranean microplate (Lowrie and Alvarez, 1974, 1975; VandenBerg, 1980a, 1980b; Edel, 1980).

The ChRM declinations were recalculated relative to their mean to compensate for regional tectonic rotation. The average inclination was used to compute the latitude of the site during deposition. Using these parameters with the appropriate inclinations, the latitude of the virtual geomagnetic pole (VGP) during deposition was calculated; it is also plotted in Figures 2 and 5. This parameter was used as an index of polarity; normal polarity corresponds to VGP latitudes near 90°N and reversed polarity to VGP latitudes near 90°S. For the Pietralata and Furlo sections, very few values are found in intermediate latitudes.

The interpreted magnetic polarity zones for the Upper Road and Pietralata are shown in Figures 2 and 5. The same parameters, determined from preliminary block samples in the Flaminia section, are shown in Figure 6; they agree with the Upper Road polarity zonation as carried across by turbidite correlation. For the Upper Cretaceous part of the Scaglia Rossa and the part of the Scaglia Bianca sampled, the polarity zone correlation to Gubbio and to the seafloor anomaly sequence (Lowrie and Alvarez, 1977a, 1977b, 1981) is straightforward (Fig. 7). The lower half of the unit has unbroken normal polarity and clearly corresponds to the Cretaceous long normal zone. Above this, all except one of the polarity zones recognized at Gubbio were found at Furlo, with their characteristic thickness fingerprint. The single exception, the very thin Gubbio C2+ subzone, may be represented by the low negative inclination at 362.3 m in the Upper Road section (Fig. 2). The Furlo Upper Road and Pietralata sections overlap from the D+ to the F+ zones, and when correlations are made on the basis of the sequence of turbidite beds (Fig. 7), the reversal levels in the two sections correspond.

Correlation of the Pietralata Paleocene is shown at expanded scale in Figure 8. Because of variable lithologies and sedimentation rates and the presence of erosional hiatuses, this correlation presented more difficulties than was the case with the Upper Cretaceous. In fact, there is no clear match of reversal fingerprints between the Paleocene sections of Pietralata and the Bottaccione section at Gubbio (Roggenthen and Napoleone, 1977; Napoleone and others, 1983). However, the Contessa Highway section at Gubbio, 2.5 km northwest of the Bottaccione (Lowrie and others, 1982), gave a fingerprint of polarity zones suggestive of a one-to-one correlation with those at Pietralata. The Contessa Highway section has been zoned using foraminifera by I. Premoli Silva (*in* Lowrie and others,

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

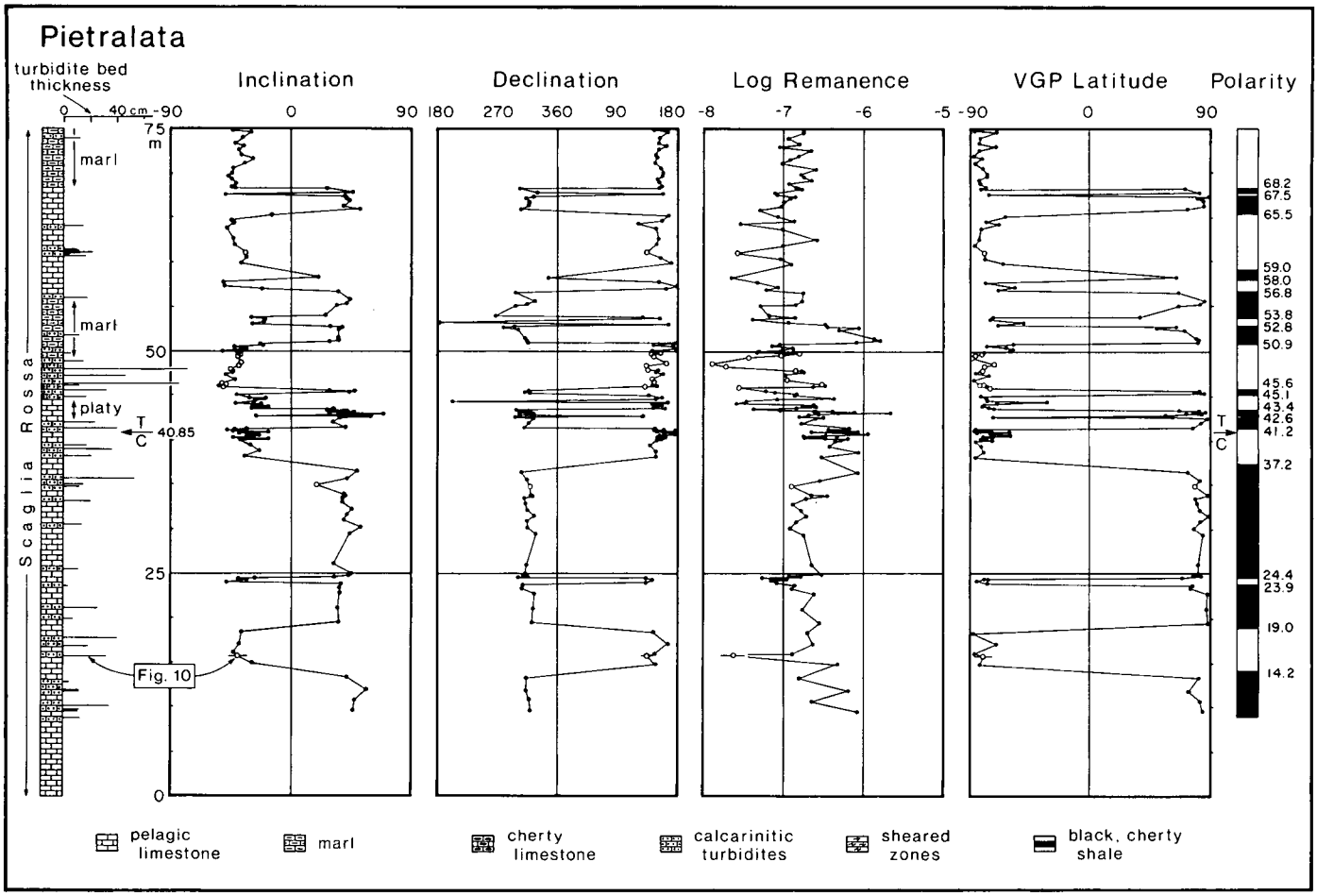


Figure 5. Magnetic results from the Pietralata section. Data points shown by open circles are from turbidite beds.

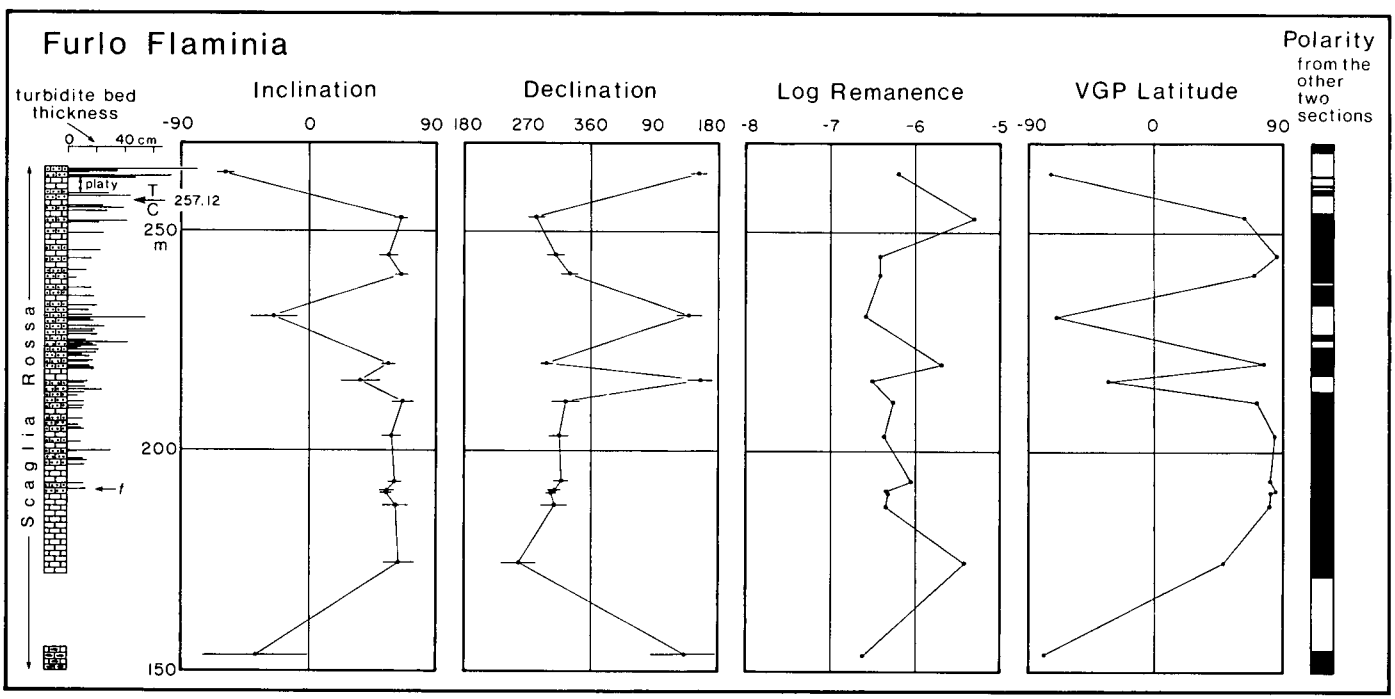


Figure 6. Magnetic results on block samples from the Furlo Flaminia section (for lithological symbols, see Fig. 5). Error bars show α_{95} (inclination) and $\alpha_{95}/\cos I$ (declination). The polarity column was *not* determined from these magnetic results; it is based on turbidite-bed correlation with the other two sections, where many more samples were measured, and shows that the few samples measured here agree with the predicted polarity zonation. The fault at 190.5 m (between the two closely spaced samples) is conspicuous in the field but does not seem to mark a major offset.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 1. PALEOMAGNETIC STATISTICS OF THE FURLO SECTIONS

Section	Number of samples	Before bedding correction				After bedding correction			
		Decl.	Incl.	α_{95}	K	Decl.	Incl.	α_{95}	K
Furlo Upper Road	204*	335	25	2.1	22.99	315	38	2.2	21.99
Furlo Flaminia	15	349	40	14.4	7.98	311	54	13.4	9.16
Pietralata	170	346	35	2.0	30.18	324	39	2.1	28.18
All samples	389	340	30	1.6	20.34	319	39	1.6	21.59
All sections	3	343	33	14.6	71.95	317	44	15.5	63.95

*Slumped intervals in Furlo Upper Road section have been omitted.

1982, and 1980, personal commun.), and although, because of poor fossil preservation, this zonation is less detailed than her previous work on the Bottaccione section (Premoli Silva, 1977; Premoli Silva and others, 1976; and *in* Napoleone and others, 1983), it allows the somewhat dissimilar reversal sequences of these two sections to be correlated with confidence (Lowrie and others, 1982). A third section at Gubbio has been measured along the Contessa Road, covering the upper Paleocene to middle Eocene. It yielded a reversal pattern that clearly correlates with the Contessa Highway and Bottaccione sections (Lowrie and others, 1982), although differences in polarity-zone thickness suggest minor stratigraphic complications at various levels in these three sections. The Paleocene of the Gubbio sections needs more work, but the pattern of polarity zones is clear enough to serve as a basis for identifying the reversal stratigraphy at Pietralata, and the information on foraminifera, although incomplete, provides support for these polarity-zone picks.

The key to the correlation is identification of the three normal intervals from 50.9 to 58.5 m at Pietralata as polarity zone 26N, which is triple in the Contessa Highway section but thinner and single in the Bottaccione section (Lowrie and others, 1982). (A note of caution: the highest normal subzone in zone 26N at Pietralata is based on a single sample at 58.22 m. On thermal demagnetization to 350 °C it shows normal polarity. At 400 and 500 °C, the intensity is too weak to be reliable, but there are suggestions of possible reversed polarity.) More work is needed to define the structure of zone 26N, as two of the normal subzones in the Contessa Highway section are also based on single samples, and marine magnetic anomalies show no evidence for short reversed intervals within anomaly 26 (J. LaBrecque and S. Cande, 1981, personal commun.). Nevertheless, the identification of zone 26N at Pietralata is confirmed by the foraminiferal identifications of A. Montanari (1980, personal commun.) and I. Premoli Silva (1981, personal commun.), which place this part of the Pietralata section in the *pseudomenardii* zone, as is the case at Gubbio. With this starting point, the remaining Pietralata polarity zones clearly tie to the Contessa Highway section and agree with the available foraminiferal information (Fig. 8).

APPLICATIONS OF POLARITY STRATIGRAPHY

The establishment of polarity-zone boundaries makes it possible to determine the average accumulation rate of compacted sediment for each polarity chron, and this information bears on sedimentological questions. Figure 9 shows the compacted accumulation rates for Furlo and Gubbio and also shows the thickness ratios of the two sections. The duration of the polarity chrons comes from interpolation in the marine magnetic-anomaly sequence (Lowrie and Alvarez, 1981). A striking feature is the reduction in sedimentation rate at about the time of the Cretaceous-Tertiary boundary, as previously noted by Arthur and Fischer (1977). This probably indicates that oceanic plankton productivity remained low throughout the Paleocene, after the shock of the terminal-Cretaceous extinctions, which are now attributed with considerable confidence to impact of an extraterrestrial object (L. W. Alvarez and others, 1980; W. Alvarez and others, 1982). The reason for the very high rates in the C- and E- zones at Gubbio is not clear. They may be artifacts, because accumulation rates of short polarity zones are more strongly affected by errors in reversal ages than are those of longer zones.

Pelagic Scaglia Rossa typically contains about 5% clay and 10% forams in a micritic matrix. When the proportions vary, as in clay-rich marl intervals, the question arises whether this is due to increase in the influx rate of one component or to diminution of the supply of the other components. This question can be resolved by the accumulation-rate information shown in Figure 9. The marl interval from 49 to 57 m at Pietralata (Fig. 8) corresponds to the abnormally high sedimentation rate in the N+ zone and thus apparently to increased clay deposition. Marl I, at 348 to 356 m (H+ to L+ zones) in the Bottaccione section (Fig. 8), is marked by accumulation rates that are much slower than those characterizing the Upper Cretaceous but are about average for the Paleocene, and, in four of the five zones, greater than at Furlo. Marl I therefore partly reflects the drop in $CaCO_3$ productivity after the terminal-Cretaceous extinction, but it may also indicate an increased clay influx. From 42.2 to 44.8 m at Pietralata (Fig. 8) is a distinctive interval of thin-bedded, platy limestones with abnormally high foram contents (40% to 75%; A. Montanari, 1980, personal commun.). The low sedimentation rates in this interval suggest removal of the fine matrix, perhaps through winnowing by bottom currents. The fine matrix removed from this interval at Furlo or similar sites may well have settled in more distal parts of the basin and thus have contributed to Marl I in sections of the Gubbio type.

Another question concerns the effect of turbidite deposition. Did the influx of turbidity-current sediment in the Furlo sections increase the rate of sediment accumulation, or did erosion by the currents remove more sediment than they deposited? Figure 9 shows that all polarity zones containing >10% turbidites are thinner at Furlo than at Gubbio. The turbidites seen in these sections evidently eroded at least enough to compensate for their added sediment contribution and commonly removed significantly more than they deposited. It should be noted, however, that erosion by turbidity currents is not the only way in which the section may be thinned. Zones I-, J+, and L+, which have no turbidites, are all thinner at Furlo than at Gubbio. This can probably be explained by noting that these three zones are marked by high foram/matrix ratios (A. Montanari, 1980, personal commun.), which we interpret as indicating scour by bottom currents.

The Bottaccione section contains three prominent marl units, 4 to 8 m thick, in the lowermost Tertiary (Fig. 8; Lowrie and others, 1982). Marl I is lower Paleocene in age, Marl II is upper Paleocene, and Marl III occurs near the base of the lower Eocene. Two prominent marl units occur in this part of the section at Pietralata, but polarity-zone correlation indicates that they do not correspond in time to any of the three marls in the Bottaccione section.

The sedimentological and stratigraphic conclusions based on the applications of magnetic stratigraphy discussed here should be considered tentative, pending further clarification of the magnetic and biostratigraphic record, but they indicate the value of paleomagnetic stratigraphy in basin-analysis studies. By determining the magnetic polarity stratigraphy of a number of Umbrian sections and combining this chronological information with sedimentological data, it should be possible to understand in considerable detail the pattern of sediment transportation and accumulation in the Umbrian basin.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

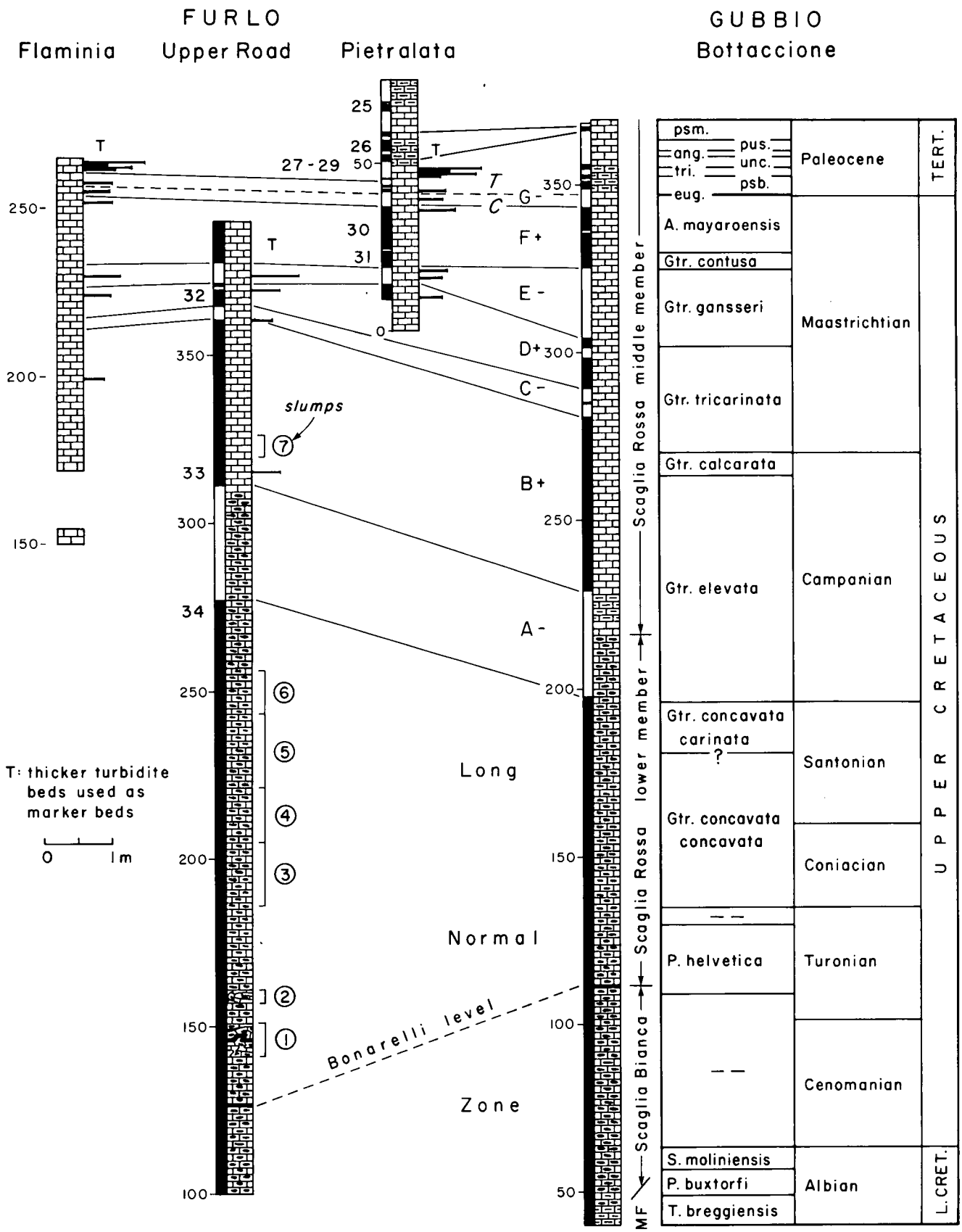


Figure 7. Correlations between the three Furlo sections and the Gubbio section, based on (1) magnetic polarity (black = normal; white = reversed; solid tie lines), (2) turbidite beds (only beds about 30 cm or thicker are shown; all turbidite beds are shown in Figs. 2, 5, and 6), (3) the Bonarelli Level (a 1-m bed of dark, siliceous shale), and (4) the Cretaceous-Tertiary boundary (T/C, based on extinction of globotruncanid foraminifera). Letters indicate the Gubbio polarity zonation (Alvarez and others, 1977). MF, fucoid Marls. Gubbio biostratigraphy after Premoli Silva and others (1976): abbreviated foram zones, from bottom up: *eugubina, pseudobulloides, trinidadensis, uncinata, angulata, pusilla pusilla, pseudomenardii.*

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

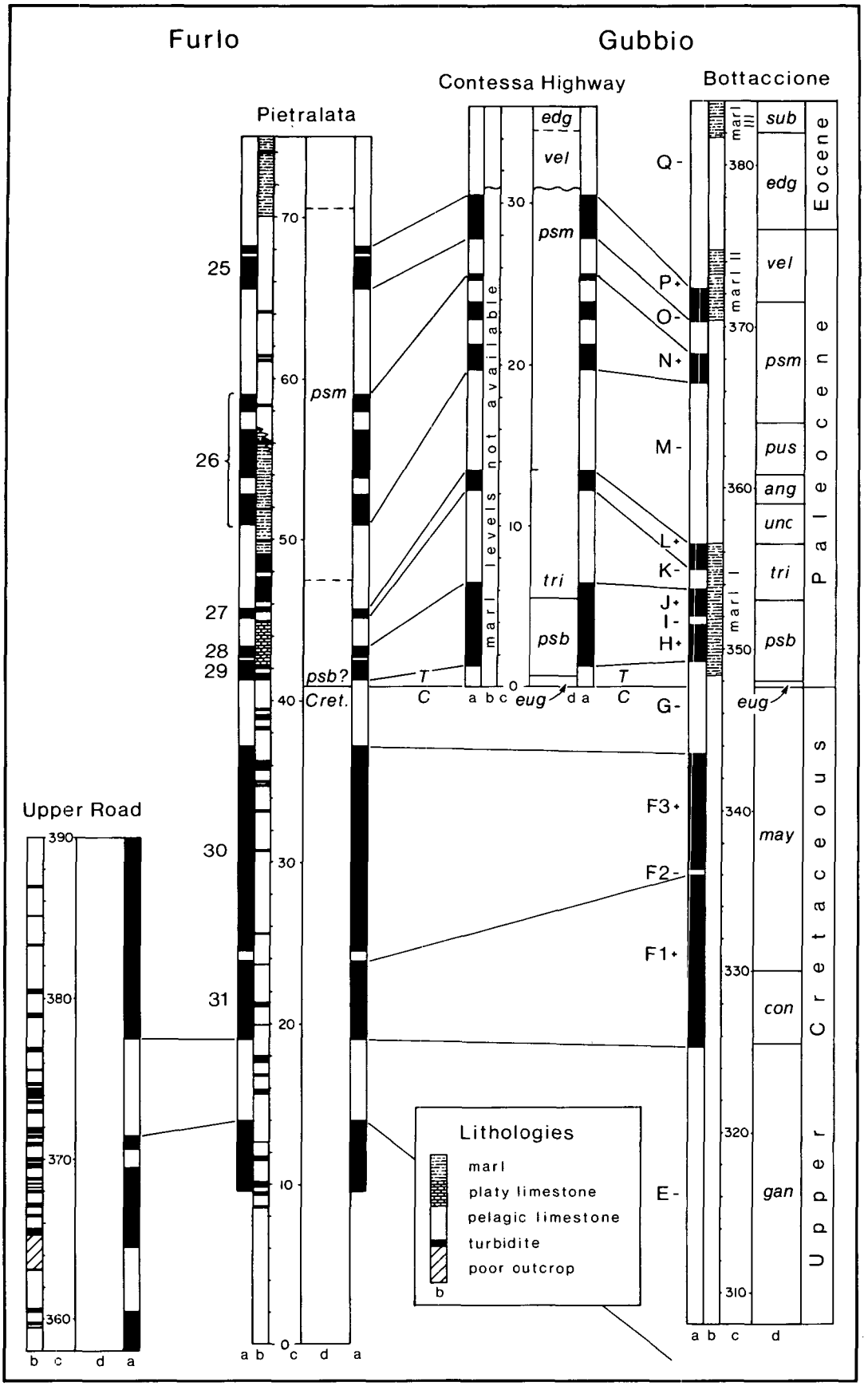


Figure 8. Correlation of Paleocene polarity zones from Pietralata to Gubbio. Columns show magnetic polarity (a), lithology (b), stratigraphic level in metres (c), and biostratigraphic zonation (d). Biostratigraphic information at Pietralata is from A. Montanari (1980, personal commun.) and I. Premoli Silva (1981, personal commun.); Contessa Highway from I. Premoli Silva (*in* Lowrie and others, 1982); Bottaccione from Premoli Silva (1977, and *in* Napoleone and others, 1983). Foram zones, from bottom up: *gansseri, contusa, mayaroensis, eugubina, pseudobulloides, trinidadensis, uncinata, angulata, pusilla pusilla, pseudomenardii, velascoensis, edgari, subbotinae.*

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

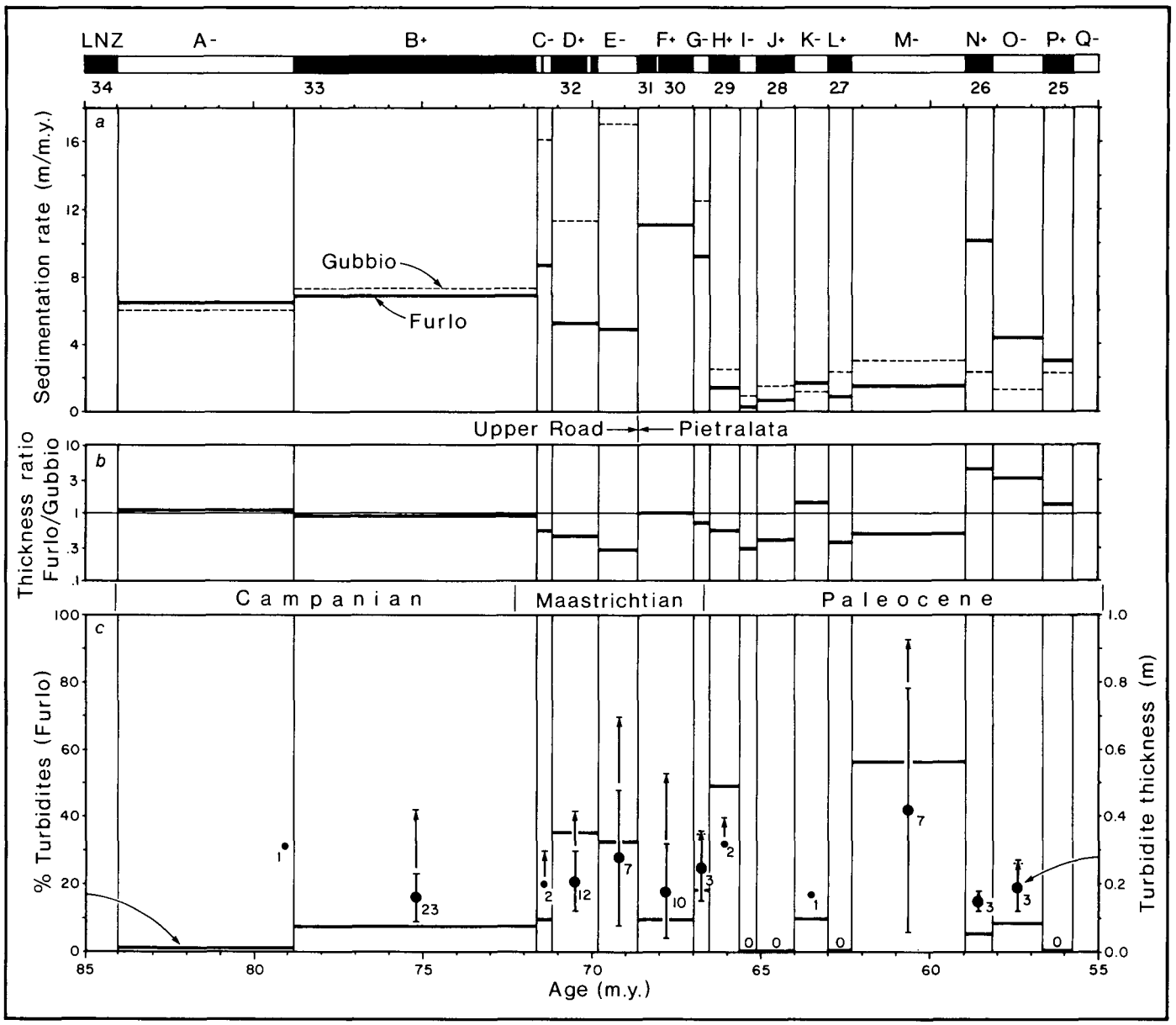


Figure 9. Sedimentation parameters plotted against age. Top: magnetic polarity zonation. a. Sedimentation rate (compacted) for Gubbio (Bottaccione) and Furlo (intervals based on the Upper Road and Pietralata sections are marked). b. Furlo/Gubbio thickness ratios. c. Furlo turbidite information: (scale at left) percent of each zone that is made up by turbidites; (scale at right) thickness of turbidite beds, showing mean, standard deviation, maximum thickness, and number of turbidite beds. Values in a and b include turbidite beds in calculated thickness and sedimentation rate.

PALEOMAGNETISM OF TURBIDITE BEDS

In the Furlo Upper Road section, we carefully avoided sampling turbidite beds. At Pietralata, on the other hand, a number of white beds, some with clear turbidite features, were sampled. Three of these samples are from the thickest turbidite in the section (at about 48 m). These beds are marked by open circles in Figure 5, and the directions obtained do not differ significantly from those measured in adjacent pink, non-turbidite beds. In addition, 12 closely spaced samples were taken spanning the turbidite bed at 15.59 to 15.91 m to examine details of remanent magnetization in a turbidite. Figure 10 shows the Bouma zonation of this bed and the magnetic results. Despite remanent intensities lower by about one order of magnitude than those of adjacent pink beds (Fig. 5), all 12 turbidite samples gave reversed directions, and the mean direction is not significantly different from those obtained from pink beds above and below.

The reliability of remanent directions in turbidite beds is to be expected in light of what is known about the mechanism of remanence acquisition in fine-grained limestones (Lowrie and Alvarez, 1975). In these rocks, extremely fine magnetite grains acquire statistical alignment with the geomagnetic field as a result of Brown-

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

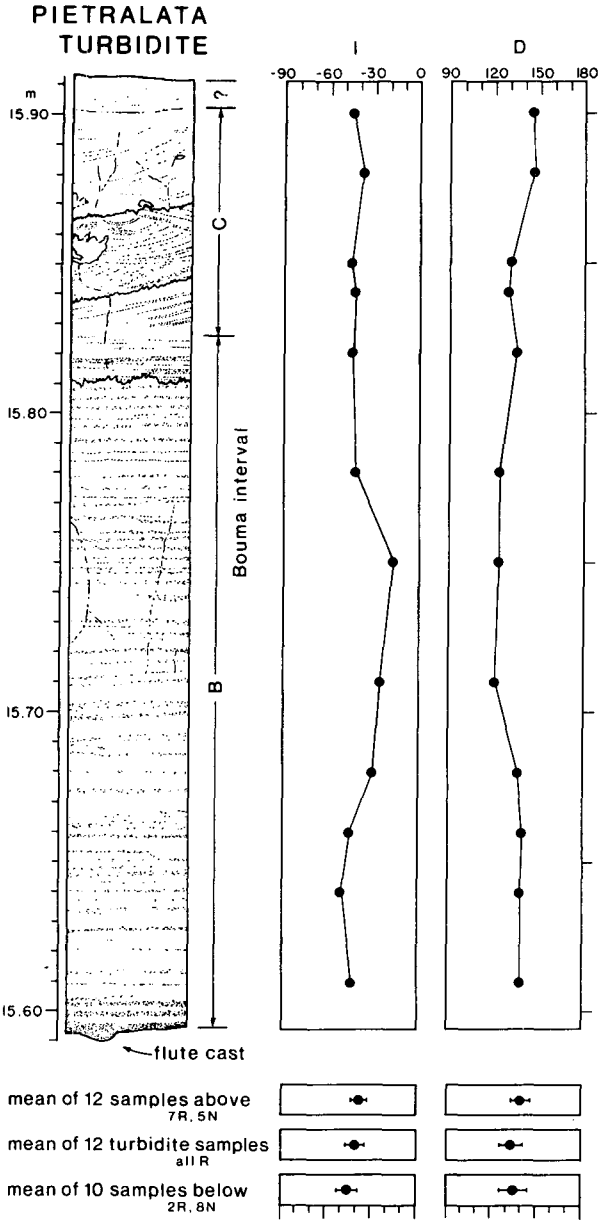


Figure 10. Detailed study of a white turbidite bed in the Pietralata section. Bedding details at left were drawn from a polished section. Inclinations, and declinations are shown at the center points of 12 samples, each 1 inch in diameter. The mean directions of the turbidite samples are indistinguishable from the mean directions of samples from the pelagic beds above and below.

ian motion in water-filled interstices, until compaction locks them into place (Irving and Major, 1964; Kent, 1973; Barton and others, 1980). The mechanism of original deposition, whether pelagic or turbiditic, does not play an important role in the acquisition of remanence by the fine-grained magnetite fraction. Wezel (1979, p. 253) raised the question "Is paleomagnetism of resedimented deposits still really valid?" and suggested that "further detailed, thorough palaeomagnetic investigations in turbidite intervals should be able to settle the validity of my heretical suggestion." Having carried out the studies recommended by Wezel, we conclude that his heretical suggestion (that the Scaglia Rossa is an entirely turbiditic unit incapable of recording the geomagnetic field) is invalid. Additional evidence contradicting Wezel's interpretation of the Scaglia Rossa has been presented by Premoli Silva and Luterbacher (1978), Premoli Silva and others (1980), and Alvarez and Lowrie (1981).

ROTATED DECLINATIONS IN SYNSEDIMENTARY SLUMPS

Disturbed zones due to soft-sediment slumping alone, or to slumping complicated by subsequent tectonic deformation, are common in the

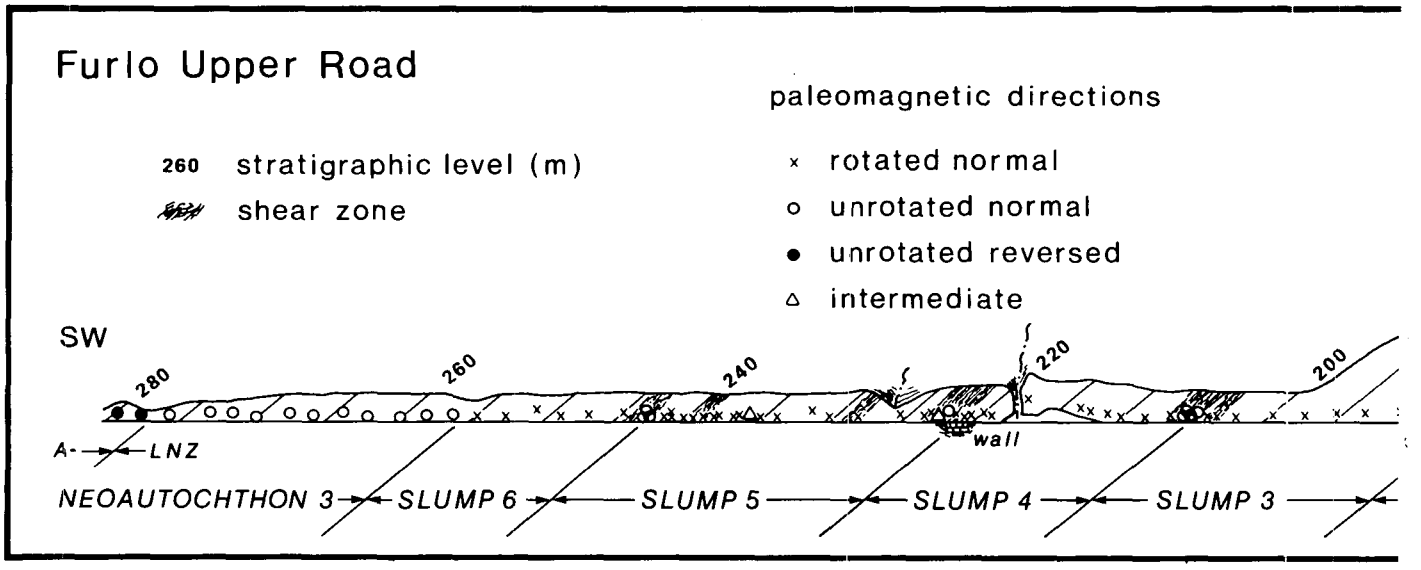


Figure 11. (below) Outcrops of the quarry (right) and road cut (left) in the lower half of the Furlo Upper Road section, showing the location and the declination characteristics of all measured samples. "Unrotated" samples have declinations close to the typical Umbrian directions—northwesterly for normal-polarity samples. Rotated directions occur in positions of the stratigraphic section that have undergone synsedimentary sliding while in a coherent condition. "Neoautochthon" refers to sediment deposited on top of a slump mass, but not itself disturbed. SR: Scaglia Rossa; SB: Scaglia Bianca.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Upper Road section below the main turbidite interval. A distinctive feature of this section, mentioned above, is the presence of the Bonarelli Level at three horizons. A particularly striking folded zone, probably resulting from a combination of both causes, occurs in the upper cherty member of the Scaglia Rossa at Pietralata, just above the interval included in this study.

What effect should soft-sediment deformation have on remanent magnetization? In view of the acquisition mechanism discussed above, the critical factor should be the degree of induration of the sediment at the time the slump takes place. If the sediment is so soft that the magnetic grains have not yet been locked into place, or if slumping remobilizes a stiff sediment on a grain-to-grain scale, the magnetic grains will be free to follow the field after the slumping occurs. The remanence eventually will be locked in when critical induration is achieved, and thus will record the ambient field sometime after slumping. If, on the other hand, the sediment remains stiff during slumping and the slump block rotates, the remanent directions will also be rotated and the magnetization should not be reset. Rotated directions thus should mark stratigraphic intervals that rotated during slumping while in a stiff condition. Unrotated directions may indicate (1) non-slumped intervals, (2) intervals that slumped while in a stiff condition but did not rotate appreciably, or (3) intervals that slumped while in a soft condition or that were liquefied during slumping and therefore had their remanent magnetization reset.

The remanent magnetization of slumped intervals in the Upper Road section (Fig. 2) supports these conjectures. Figure 11 shows the part of the Upper Road section that contains the main slump intervals, with all magnetic samples marked as showing rotated or unrotated directions. The following intervals are identified:

Autochthon. ("Autochthon" is used here for the part of a local sedimentary sequence that predates the slumping episodes; it is not intended to imply autochthony of the Umbrian sequence.) Normal stratigraphic sequence of Upper Scaglia Bianca, Bonarelli Level I, and basal Scaglia Rossa; no disturbed bedding; all declinations unrotated.

Slump 1. This interval probably represents a single slump event, but it contains five recognizable units:

(a) Scaglia Bianca with strongly disturbed bedding and reset magnetic directions; four unrotated directions occur in smoothly curving folds and thus probably represent magnetic resetting; two rotated directions seem to be in blocky bed fragments that may have been carried along in the softer slump matrix without losing their stiffness, but this is uncertain.

(b) The top three beds of the Scaglia Bianca plus Bonarelli Level II, showing only minor internal disruption, but thrown into a fold with overturn indicating northeastward motion; Bonarelli Level II is shown to be a structural repetition of Bonarelli Level I by identical thicknesses and by identical stratigraphic details within and below the two levels.

(c) Probably basal Scaglia Rossa, with undisturbed bedding and rotated declinations; it is probable that this unit, Bonarelli Level II, and the three undisturbed beds at the top of the Scaglia Bianca rode as a coherent package over the strongly disturbed unit at the base of the slump.

(d) Bonarelli Level III, abnormally thin, strongly deformed internally, and variable in thickness.

(e) White limestone, strongly deformed internally, with one unrotated direction: units d and e probably represent soft material that spilled down over the top of the coherent slab.

Neoautochthon 1. ("Neoautochthon" is a term used in the Italian tectonic literature for sedimentary rocks that have not been transported laterally but that were deposited on top of a nappe or thrust sheet after its emplacement under water. Again, it is used here in a local sense only.) Lower member Scaglia Rossa with typical pink chert, undisturbed bedding, and unrotated directions; this interval was probably deposited by normal pelagic processes after emplacement of slump 1 and never moved. (All intervals from here to top of section in Figure 11 are Scaglia Rossa, lower cherty member.)

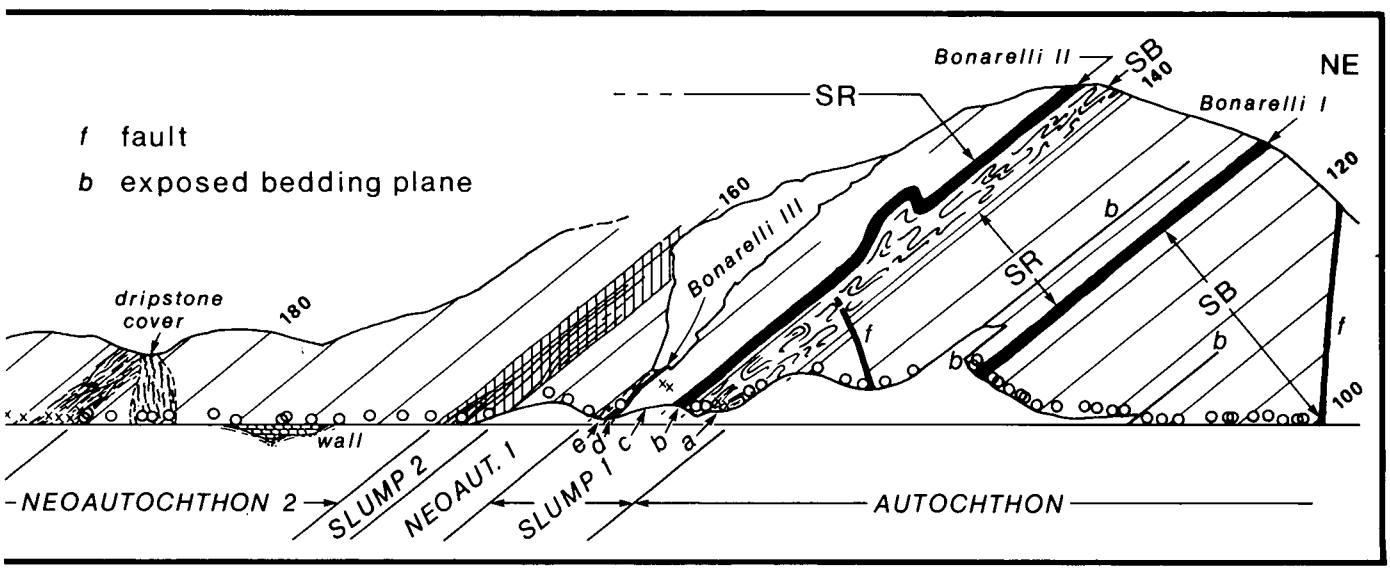
Slump 2. This interval shows long, recumbent, synclinal folds; tight but smoothly curving fold hinges; clearly secondary thickness changes in limestone beds; and unrotated declinations. It represents a slump of material that was soft enough for the magnetization to be reset.

Neoautochthon 2. Similar to neoautochthon 1 (could perhaps be a slump that did not rotate).

Slump 3. This 15-m sequence of undisturbed beds with rotated directions represents a coherent slump block, the motion of which was accommodated by deformation in the 3-m shear zone at its base. Apparently, only the very bottom of the shear zone was disturbed enough to have its magnetization reset.

Slumps 4, 5, and 6. Similar in thickness and features to slump 3. The thin shear zones within slump 5 are of uncertain significance.

Figure 11. (*Continued*).



 Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 2. AVERAGE DIRECTIONS IN SLUMPED BLOCKS, FURLO UPPER ROAD SECTION

Slump	Block limits (m)	Number of samples	Before bedding correction				After bedding correction*			
			Decl.	Incl.	α_{95}	K	Decl.	Incl.	α_{95}	K
6	255 245	6	27.9	10.2	8.5	63.48	24.0	49.3	6.4	111.4
5	242 223	19	15.4	0.7	5.2	42.44	11.0	34.0	5.2	43.07
4	221.5 206	11	35.8	4.7	6.8	45.87	36.3	38.4	7.1	42.78
3	204.5 186	13	31.7	7.2	8.3	25.65	30.7	32.1	7.3	32.89
Average for slumped blocks		6	24.2	0.7	9.4	52.03	21.3	38.2	9.2	53.93
Samples between slumps	255 186	11	340.3	20.9	8.6	29.33	320.2	33.4	10.6	19.38

*Tectonic corrections made by simple rotation about present strike line.

Neoautochthon 3. The base of this unit is in a continuous sequence of undisturbed beds and is marked only by the change from rotated to unrotated declinations. Neoautochthon 4 continues upward to the slump interval at 319.4–326.0 m, which marks the beginning of the main turbidite sequence (Fig. 2).

In a study of the Sicilian Scaglia, Channell and others (1980, Fig. 8) concluded that remanence was acquired after synsedimentary folding but before tectonic folding. Results from Furlo are in agreement but also show that remanence acquisition predated movement of coherent synsedimentary slide masses.

We do not know why slumps are concentrated in the Furlo area, but they agree with the abnormal stratigraphic reduction of Jurassic to Lower Cretaceous units in suggesting unusual synsedimentary mobility of this area. It is peculiar that slumps 3 to 6 are all about the same thickness and are rotated approximately the same amount in the same direction (Table 2). Unrotated directions in the shear zones at the base of each slump show that the four slumps were not first stacked up and then rotated together; evidently, each slump mass was emplaced individually with the same rotation. Slumps 3 to 6 may represent the imbricated front of a single coherent sheet which rotated during movement as a unit, with imbricate stacking occurring as the last part of the slumping event.

ACKNOWLEDGMENTS

We thank Milly Alvarez, Marcia Lowrie, Corrado Concetti, Fausto Pellici, Stephen Marshak, and William S. Leith for help with the field work and Wolfgang Gruber for his careful laboratory measurements. We are indebted to F. Medizza, A. Micarelli, A. Montanari, and I. Premoli Silva for foraminiferal determinations. Peter A. Geiser, Alessandro Montanari, Isabella Premoli Silva, and Alfred G. Fischer helped us during discussions of the stratigraphy and structure of Furlo. Peter N. Shive and Sue Beske-Diehl offered useful comments in reviews for the journal. This work was partially supported by grants EAR-76-03272, EAR-79-11337, and EAR-80-22846 from the National Science Foundation, Geology Program, to Alvarez. This paper is Institut für Geophysik, ETH-Zürich, Contribution No. 332.

REFERENCES CITED

Alvarez, L. W., Alvarez, W., Asaro, F., and Michel, H. V., 1980, Extraterrestrial cause for the Cretaceous-Tertiary extinction: Science, v. 208, p. 1095–1108.

Alvarez, W., and Lowrie, W., 1978, Upper Cretaceous paleomagnetic stratigraphy at Moria (Umbrian Apennines, Italy): Verification of the Gubbio section: Geophysical Journal of the Royal Astronomical Society, v. 55, p. 1–17.

——1981, Upper Cretaceous to Eocene pelagic limestones of the Scaglia Rossa are not Miocene turbidites: Nature, v. 294, p. 246–248.

Alvarez, W., Arthur, M. A., Fischer, A. G., Lowrie, W., Napoleone, G., Premoli Silva, I., and Roggenthen, W. M., 1977, Upper Cretaceous Paleocene magnetic stratigraphy at Gubbio, Italy—V, Type section for the Late Cretaceous Paleocene geomagnetic reversal time scale: Geological Society of America Bulletin, v. 88, p. 383–389.

Alvarez, W., Alvarez, L. W., Asaro, F., and Michel, H. V., 1982, Current status of the impact theory for the terminal Cretaceous extinction, *in* Geological implications of impacts of large asteroids and comets on the Earth: Geological Society of America Special Paper 190, p. 305–315.

Arthur, M. A., and Fischer, A. G., 1977, Upper Cretaceous-Paleocene magnetic stratigraphy at Gubbio, Italy—I, Lithostratigraphy and sedimentology: Geological Society of America Bulletin, v. 88, p. 367–371.

Barton, C. E., McElhinny, M. W., and Edwards, D. J., 1980, Laboratory studies of depositional DRM: Royal Astronomical Society Geophysical Journal, v. 61, p. 355–377.

Channell, J.E.T., 1976, Umbrian paleomagnetism and the concept of the African-Adriatic Promontory: Società Geologica Italiana Bollettino, v. 15, p. 119–128.

——1978, Dual magnetic polarity measured in a single bed of Cretaceous pelagic limestone from Sicily: Journal of Geophysics, v. 44, p. 613–622.

Channell, J.E.T., and Horvath, F., 1976, The African/Adriatic Promontory as a paleogeographical premise for Alpine orogeny and plate movements in the Carpatho-Balkan region: Tectonophysics, v. 35, p. 71–101.

Channell, J.E.T., and Tarling, D. H., 1975, Palaeomagnetism and the rotation of Italy: Earth and Planetary Science Letters, v. 25, p. 177–188.

Channell, J.E.T., Lowrie, W., Medizza, F., and Alvarez, W., 1978, Palaeomagnetism and tectonics in Umbria, Italy: Earth and Planetary Science Letters, v. 39, p. 199–210.

Channell, J.E.T., D'Argenio, B., and Horvath, F., 1979, Adria, the African promontory, in Mesozoic Mediterranean palaeogeography: Earth Science Reviews, v. 15, p. 213–292.

Channell, J.E.T., Catalano, R., and D'Argenio, B., 1980, Palaeomagnetism and deformation of the Mesozoic continental margin in Sicily: Tectonophysics, v. 61, p. 391–407.

Channell, J.E.T., Freeman, R., Heller, F., and Lowrie, W., 1982, Timing of diagenetic haematite growth in red pelagic limestones from Gubbio (Italy): Earth and Planetary Science Letters, v. 58, p. 189–201.

Edel, J. B., 1980, Étude paléomagnetique de la Sardaigne: Consequences pour la géodynamique de la Méditerranée occidentale [dissert.]: Strasbourg, University Louis Pasteur.

Fisher, R. A., 1953, Dispersion on a sphere: Royal Society of London Proceedings, ser. A, v. 217, p. 295–305.

Hamano, Y., 1980, An experiment on the post-depositional remanent magnetization in artificial and natural sediments: Earth and Planetary Science Letters, v. 51, p. 221–232.

Irving, E., 1957, The origin of the palaeomagnetism of the Torridonian Sandstone series of northwest Scotland: Royal Society of London Philosophical Transactions, ser. A, v. 250, p. 100–110.

Irving, E., and Major, A., 1964, Post-depositional detrital remanent magnetization in a synthetic sediment: Sedimentology, v. 3, p. 135–143.

Kent, D. V., 1973, Post-depositional remanent magnetization in deep-sea sediment: Nature, v. 246, p. 32–34.

Lowrie, W., and Alvarez, W., 1974, Rotation of the Italian Peninsula: Nature, v. 251, p. 285–288.

——1975, Paleomagnetic evidence for rotation of the Italian Peninsula: Journal of Geophysical Research, v. 80, p. 1579–1592.

——1977a, Upper Cretaceous–Paleocene magnetic stratigraphy at Gubbio, Italy—III, Upper Cretaceous magnetic stratigraphy: Geological Society of America Bulletin, v. 88, p. 374–377.

——1977b, Late Cretaceous geomagnetic polarity sequence: Detailed rock and paleomagnetic studies of the Scaglia Rossa limestone at Gubbio, Italy: Royal Astronomical Society Geophysical Journal, v. 51, p. 561–581.

——1981, One hundred million years of geomagnetic polarity history: Geology, v. 9, p. 392–397.

Lowrie, W., and Heller, F., 1982, Magnetic properties of marine limestones: Reviews of Geophysics and Space Physics, v. 20, p. 171–192.

Lowrie, W., Alvarez, W., Premoli Silva, I., and Monechi, S., 1980a, Lower Cretaceous magnetic stratigraphy in Umbrian pelagic carbonate rocks: Royal Astronomical Society, Geophysical Journal, v. 60, p. 263–381.

Lowrie, W., Channell, J.E.T., and Alvarez, W., 1980b, A review of magnetic stratigraphy investigations in Cretaceous pelagic carbonate rocks: Journal of Geophysical Research, v. 85 (Normal Watkins Memorial Issue), p. 3597–3605.

Lowrie, W., Alvarez, W., Napoleone, G., Perch-Nielsen, K., Premoli Silva, I., and Toumarkine, M., 1982, Paleogene magnetic stratigraphy in Umbrian pelagic carbonate rocks: The Contessa sections, Gubbio: Geological Society of America Bulletin, v. 93, p. 414–432.

Montanari, A., 1979, Lineamenti sedimentologici della "Scaglia Bianca" e della "Scaglia Rossa" nelle Marche settentrionali: Laurea Thesis, University of Urbino, Italy, 289 p.

Napoleone, G., 1980, Lo sviluppo della stratigrafia magnetica nella sezione tipo di Gubbio e sua incidenza nella ricerca paleomagnetica: Società Geologica Italiana Memorie, v. 21, p. 289–299.

Napoleone, G., Premoli Silva, I., Heller, F., Cheli, P., Corezzi S., and Fischer, A. G., 1983, Eocene magnetic stratigraphy at Gubbio, Italy, and its implication for Paleogene geochronology: Geological Society of America Bulletin, v. 94, p. 181–191.

Premoli Silva, I., 1977, Upper Cretaceous–Paleocene magnetic stratigraphy at Gubbio, Italy—II, Biostratigraphy: Geological Society of America Bulletin, v. 88, p. 371–374.

Premoli Silva, I., and Luterbacher, H. P., 1978, The alleged Miocene age of the "Scaglia Rossa" in the Bottaccione section near Gubbio (central Italy): Rivista Italiana di Paleontologia e Stratigrafia, v. 84, p. 667–672.

Premoli Silva, I., Napoleone, G., and Fischer, A. G., 1974, Risultati preliminari sulla stratigrafia paleomagnetica della Scaglia cretaceo-paleocenica della sezione di Gubbio (Appennino centrale): Società Geologica Italiana Bollettino, v. 93, p. 647–659.

Premoli Silva, I., Paggi, L., and Monechi, S., 1976, Cretaceous through Paleocene biostratigraphy of the pelagic sequence at Gubbio, Italy: Società Geologica Italiana Memorie, v. 15, p. 21–32.

Premoli Silva, I., Napoleone, G., and Fischer, A. G., 1980, La sezione magnetostratigrafica di Gubbio: Indagini nella storia del Cretacico-Paleogene: Società Geologica Italiana Memorie, v. 21, p. 301–311.

Roggenthen, W. M., and Napoleone, G., 1977, Upper Cretaceous–Paleocene magnetic stratigraphy at Gubbio, Italy—IV, Upper Maastrichtian–Paleocene magnetic stratigraphy: Geological Society of America Bulletin, v. 88, p. 378–382.

Tucker, P., 1980, Stirred remanent magnetization: A laboratory analogue of post-depositional realignment: Journal of Geophysics, v. 48, p. 153–157.

VandenBerg, J., 1980a, Reconstructions of the western Mediterranean area for the Mesozoic and Tertiary timespan: Geologie en Mijnbouw, v. 58, p. 153–160.

——1980b, Palaeomagnetic data from the western Mediterranean: A review: Geologie en Mijnbouw, v. 58, p. 161–174.

VandenBerg, J., Klootwijk, C. T., and Wonders, A.A.H., 1978, Late Mesozoic and Cenozoic movements of the Italian Peninsula: Further palaeomagnetic data from the Umbrian sequence: Geological Society of America Bulletin, v. 89, p. 133–150.

Wezel, F. C., 1979, The Scaglia Rossa formation of central Italy: Results and problems emerging from a regional study: L'Atenec Parmense Acta Naturalia, v. 15, p. 243–259.

MANUSCRIPT RECEIVED BY THE SOCIETY MAY 27, 1982
REVISED MANUSCRIPT RECEIVED DECEMBER 2, 1982
MANUSCRIPT ACCEPTED DECEMBER 22, 1982